MOHAMMAD ORABE

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I am an applied machine learning researcher with interests in representation learning, probabilistic modeling, uncertainty quantification, and learning from heterogeneous data sources. My work combines deep learning and statistical inference to develop robust learning methods for complex data, with applications in neuroscience, biomedical sensing, and scientific machine learning. I am interested in understanding how learned representations generalize across domains and how their structure can be leveraged to improve robustness, adaptation, and interpretability.

EDUCATION

M.Sc. COMPUTER SCIENCE – TU Berlin — Apr 2022 – Mar 2026

- Focus: Probabilistic Modeling, Deep Learning, Computer Vision, Signal Analysis, Human-Computer Interaction.
- Thesis: Deep Learning-Based Classification of Gait Modulation from Local Field Potentials. Thesis | Code

B.Sc. COMPUTER SCIENCE – TU Berlin — Oct 2018 – Mar 2022

- Focus: Scientific Computing, Probability & Statistics, Optimization, Computer Vision, Cognitive Systems.
- Thesis: Markerless Pose Estimation and Mice Tracking with Deep Learning. Thesis | Code

RESEARCH EXPERIENCE

MACHINE LEARNING RESEARCHER (FULL-TIME) — Apr 2026 – Present
BIFOLD | Charité | BMW GROUP

- Conducted research within the AutoHealth project, a collaboration between BIFOLD, Charité, and BMW Group focused on AI-driven in-car cardiovascular monitoring and physiological sensing in real-world driving environments.
- Designed dynamic physiological feature extraction and multimodal analysis methods for stress-response modeling and cardiovascular risk assessment under controlled and real-world driving conditions.

RESEARCH STUDENT ASSISTANT (PART-TIME) — Feb 2025 – Mar 2026
PTB Berlin | Quality in AI Labs | Supervisor: Prof. Stefan Haufe — Project

- Conduct research on uncertainty quantification and calibration in inverse problems for M/EEG source imaging.
- Developed a benchmarking framework to evaluate posterior uncertainty and calibration of Bayesian estimators.

RESEARCH STUDENT ASSISTANT — Feb 2024 – Mar 2026
TU Berlin | Quality in AI Labs | UNIML Department | Supervisor: Prof. Stefan Haufe — Project

- Develop CNN-, LSTM-, and distillation-based models for gait impairment decoding from multimodal biosignals.
- Design and evaluate training pipelines for real-time constraints, focusing on cross-subject generalization.

RESEARCH STUDENT ASSISTANT (PART-TIME) — Jul 2023 – Dec 2024
Charité | Brain & Data Science Lab | Supervisor: Prof. Stefan Haufe — Tutorial 1 | Tutorial 2

- Investigated directed and frequency-domain interactions in electrophysiological time-series data.
- Extended multivariate connectivity analysis functionality in the widely used open-source scientific toolbox MNE.

RESEARCH STUDENT ASSISTANT — Jul 2021 – Jun 2023
HU Berlin | WinterLab | Supervisor: Prof. York Winter — Project 1 | Project 2 | Project 3

- Applied deep learning methods to analyze social behavior and interaction patterns in mice from behavioral data.
- Developed transfer learning- and CNN-based computer vision pipelines for motion tracking and object detection.

RESEARCH SCHOLARSHIPS

RESEARCH STIPEND IN DEEP LEARNING — Sep 2025 – Present
BIFOLD | IBS Lab | Supervisor: Dr. Alexander von Luehmann — Project

- Awarded research funding to investigate data augmentation, transfer learning approaches for adapting deep learning models across heterogeneous sensing configurations and varying data densities in fNIRS.

PUBLICATIONS & CONFERENCE CONTRIBUTIONS

Orabe, M., Isaias, I., & Haufe, S. Deep Learning-Based Classification of Gait Modulation from Local Field Potentials. Manuscript in preparation (first author), 2026.

Orabe, M., Haufe, S., & Huseynov, I. CaliBrain: A Framework for Calibrated Uncertainty Quantification in M/EEG Source Imaging. Manuscript in preparation for *Journal of Open Source Software*, 2026. Docs

Orabe, M., Dissanayake, T., Moradi, S., & von Lühmann, A. Deep Learning Modeling for Sparse-to-High-Density fNIRS Conversion. Mobile Brain/Body Imaging Conference, 2026 (abstract submitted and accepted). Abstract

Huseynov, I., **Orabe, M.**, Hashemi, A., Nagarajan, S., & Haufe, S. Establishing a Framework to Measure Uncertainty Calibration in M/EEG Source Imaging. Manuscript in preparation, 2026. Poster.

Binns, T. S., **Orabe, M.**, ..., Haufe, S. (2024). Multivariate Connectivity Methods in the MNE-Python Toolbox. *Neural Traces*, Berlin, Germany. Poster.

GRADUATE PROJECTS (SELECTED)

MULTI-MODAL GENERATIVE MODELING – TU Berlin | Report | Code

Investigated multimodal generative models to learn joint latent representations from images and text, focusing on robust cross-modal inference and weakly supervised learning under missing modalities.

IMAGE SEGMENTATION AND MOLECULAR PREDICTION – TU Berlin | Report | Code

Investigated deep learning approaches for brain tumor sub-region segmentation and prediction of clinically relevant molecular biomarkers from multi-parametric MRI data.

BAYESIAN MODELING OF MULTISENSORY PERCEPTION – TU Berlin | Report | Code

Studied Bayesian causal inference models for multisensory integration of visual and auditory signals under uncertainty, and evaluated model behavior through simulation-based experiments.

Neural Simulation Framework (neurolib) – TU Berlin | Poster | Report | Code

Extended an open-source library (neurolib) for simulating brain activity based on anatomical models.

TEACHING & SCIENTIFIC SERVICE

FOUNDER & LEAD INSTRUCTOR – MathCodeLab Berlin | Website | Aug 2023 – Present

- Founded and led a STEM-focused education academy offering in-person and online courses in math & CS.
- Designed and led 15+ intensive 12-week STEM university-level academic tutoring courses, building and mentoring a cohort of 150+ undergraduate students across Germany and Denmark.

TEACHING ASSISTANT – TU Berlin | UNIML Department | Oct 2024 – Apr 2025

- Assisted in a Master's-level seminar focused on advanced machine learning for biomedical data science.
- Supported students in literature analysis, methodological discussion, and scientific presentation.

PROGRAM COMMITTEE MEMBER – PTB | Neural Traces Conference | Apr 2024

- Contributed to the scientific process for a conference on advanced M/EEG methods and clinical applications.

TECHNICAL SKILLS

- **Programming Languages:** Python, R, MATLAB, Java, C/C++
- **ML Frameworks:** TensorFlow, PyTorch, Keras, scikit-learn, Hugging Face Transformers, Pandas, SciPy, OpenCV
- **Cloud & Infrastructure:** PostgreSQL, Render, Railway, Docker, CI/CD
- **Development:** Git, Bash, SLURM, MLOps
- **Soft Skills:** Self-motivation, clear communication, independence & collaborative teamwork

OPEN-SOURCE RESEARCH SOFTWARE (LEAD DEVELOPER)

CaliBrain (Python) – PTB | Code

Lead developer of a benchmarking framework for uncertainty quantification and calibration in M/EEG source imaging with systematic evaluation of posterior uncertainty behavior and statistical calibration of Bayesian inverse models.

GaitMod (Python) – PTB | Charité Docs
Developed a unified ML framework with full analytical pipeline: signal preprocessing, feature engineering and selection, statistical analysis, comprehensive suite of deep learning architectures, and hyperparameter optimization.

GroupMouseTrack (Python) – WinterLab Code
Designed and implemented a deep learning framework combining computer vision and RFID sensing for robust pose estimation and identity-preserving multi-animal tracking in longitudinal rodent behavioral studies.

ColonyRack (R) – WinterLab Code
Built a toolbox for large-scale processing and statistical analysis of tabular and time-series behavioral data from laboratory mice to enable extraction of activity patterns and social interaction metrics.

INDUSTRY EXPERIENCE & INTERNSHIPS

IoT MIDDLEWARE DEVELOPER – DAI Lab | GT-ARC Berlin Jul 2020 – Oct 2020

- Designed and implemented middleware components for interoperable communication across IoT devices.

IT SYSTEMS ASSISTANT – Institute for International Communication - Berlin Feb 2017 – Jul 2020

- Maintained operating systems; supported internal network and user infrastructure.
- Diagnosed and resolved software issues; supported automation and optimization of internal technical workflows.

REFERENCES

Prof. Dr. Stefan Haufe, Head of UNIML (TU Berlin), ML & Uncertainty (PTB), and Braindata (Charité Berlin).
Prof. Dr. Srikantan Nagarajan, Head of Radiology and Biomedical Imaging, University of California, San Francisco.
Dr. Alexander von Luehmann, Head of Intelligent Biomedical Sensing research group, TU Berlin and BIFOLD.
Dr. Ismail Huseynov, Senior research associate at PTB.
Dr. Ali Hashemi, (Former) Senior research scientist at BIFOLD.
Dr. Theekshana Dissanayake, Post-Doctoral researcher at IBS research group / BIFOLD.

LANGUAGES

English: Proficient (C1)
German: Proficient (C1)
Arabic: Native